

RISK & RETURN IN INVESTING

Financial Mathematics · CA-III · Unit V and Unit VI

Name: _____

PRN / Roll No: _____

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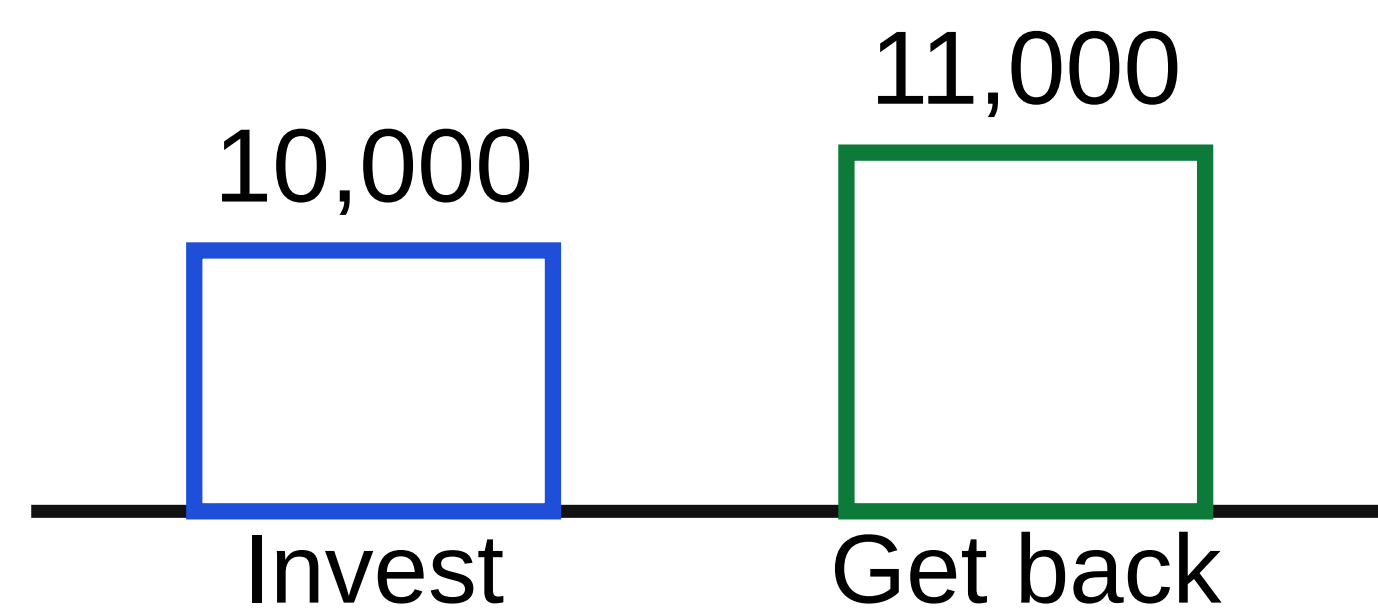
UNIT V: BASICS OF INVESTMENT

1. Investment Return

The gain on an investment, as a % of money invested.

$$\text{Return} = (\text{Gain} \div \text{Invested}) \times 100$$

Eg: Invest ₹10,000, get back ₹11,000.
Return = $1,000 \div 10,000 = 10\%$

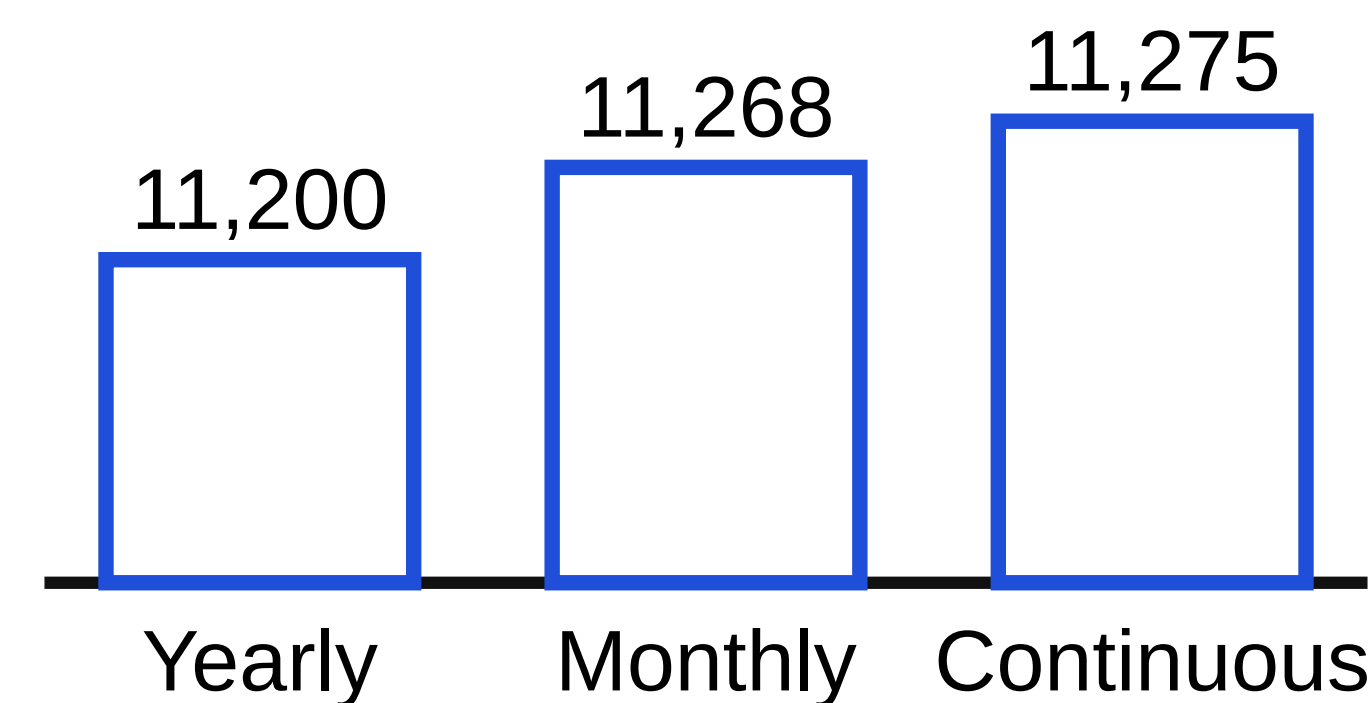


2. Compounding Frequency

More times interest is added per year = more money.

$$A = P (1 + r/m)^{mt}$$

Eg: ₹10,000 at 12% for 1 year



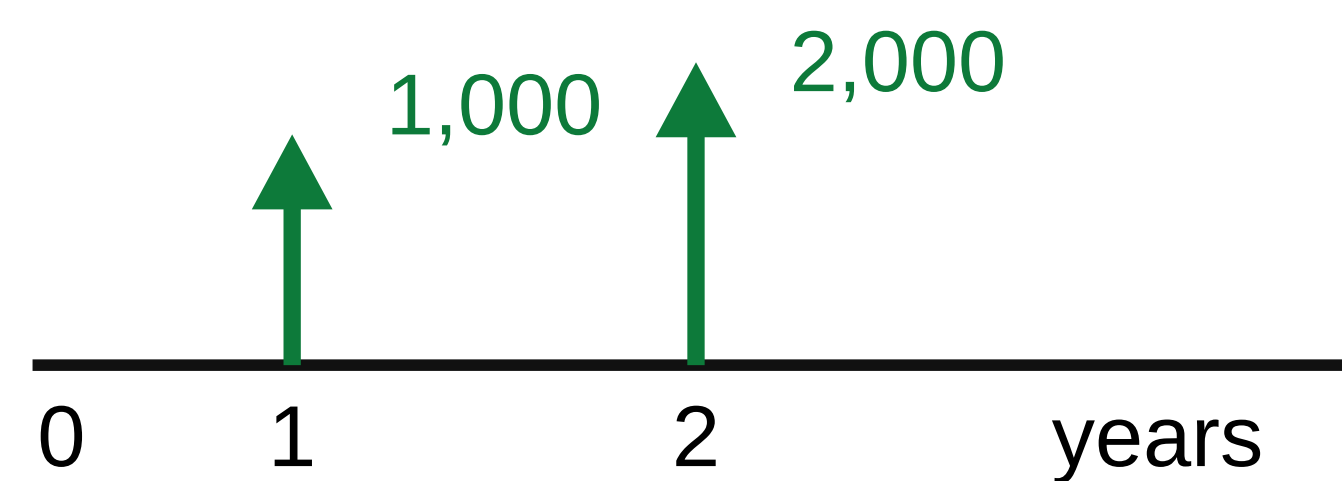
3. Uneven Cash Flows

Money received is different each year. Bring each back to today.

$$PV = CF_1/(1+i) + CF_2/(1+i)^2$$

Eg: ₹1,000 (yr 1) and ₹2,000 (yr 2), $i = 10\%$

$$PV = 909 + 1,653 = ₹2,562$$

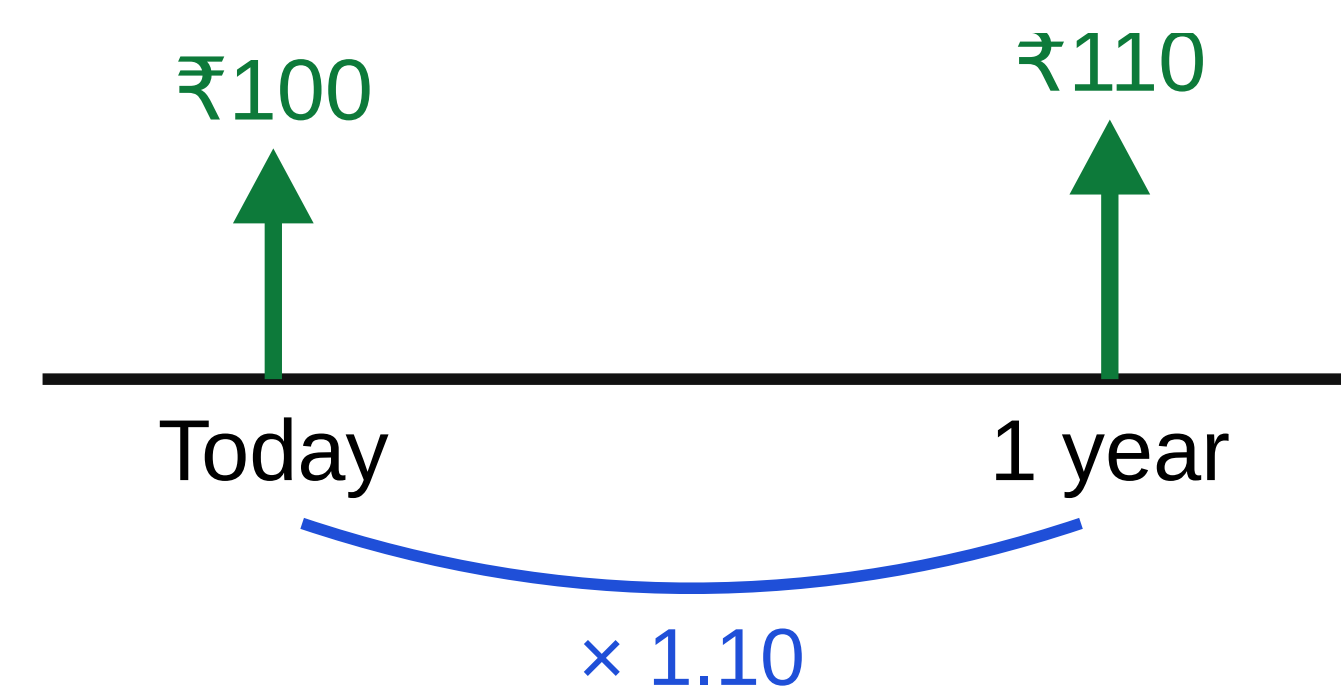


4. Economic Equivalence

Different amounts at different times can be worth the same.

$$F = P (1 + i)^n$$

Eg: at 10%, ₹100 today = ₹110 after 1 year



UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk

Risk = chances known. Pick the best average (EMV).

$$EMV = \Sigma (\text{chance} \times \text{payoff})$$

50%: ₹100, 50%: ₹40 $\Rightarrow EMV = ₹70$



2. Decision under Uncertainty

Chances unknown. Use a rule:

- **Maximax**: best of the best
- **Maximin**: best of the worst

Eg: Stock (80 or -20), Bond (40 or 20)

Maximax → Stock

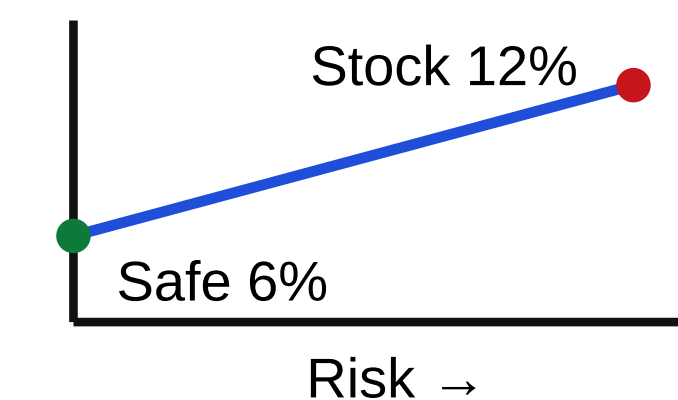
Maximin → Bond

3. Risk Premium

Extra return for taking risk.

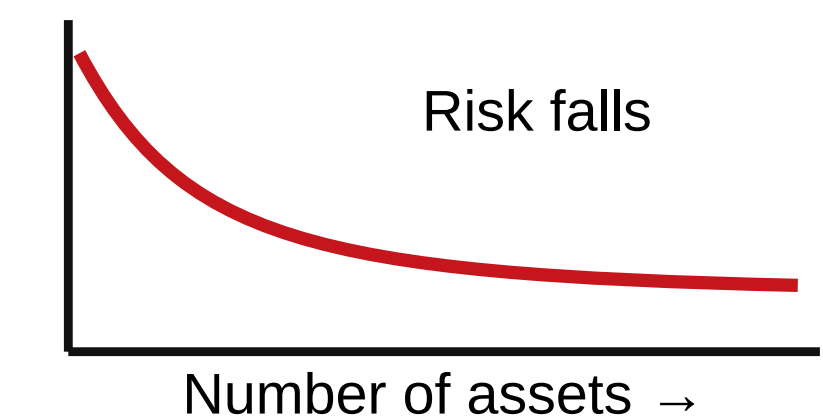
$$\text{Premium} = \text{Expected} - \text{Safe rate}$$

Eg: $12\% - 6\% = 6\%$



4. Portfolio Diversification

Don't put all eggs in one basket: more different assets = less risk.



5. Life Insurance

You pay a premium; the company pays the sum assured if you die.

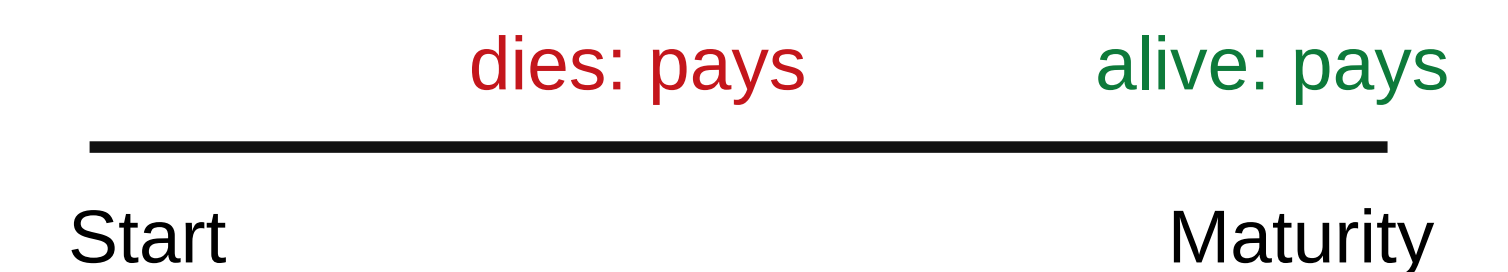
$$\text{Premium} \approx \text{Sum assured} \times P(\text{death})$$

Eg: $₹1,00,000 \times 0.01 = ₹1,000$

6. Endowment

Pays if you die in the term OR survive to the end.

$$\text{Endowment} = \text{Insurance} + \text{Savings}$$



Higher risk $\Rightarrow$ higher expected return